MIL-C-21665B(SHIPS)
13 August 1965
SUPERSEDING
MIL-C-21665A(SHIPS)
12 April 1965
(See 6.6)

MILITARY SPECIFICATION

CATALYST, CARBON MONOXIDE AND HYDROCARBON OXYDIZING (FOR USE IN AN AIR PURIFYING DEVICE ABOARD SUBMARINES)

1. SCOPE

1.1 This specification covers one grade of catalyst for atmospheric air purification.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves: Standard For Testing Purposes.
PPP-P-704 - Pails; Shipping, Steel (1 Through 12 Gallon).

MILITARY

MIL-L-20213 - Lithium Hydroxide (LiOH), Technical.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" X 48" 4-Wa (Partial) Pallet Skids, Runners, Or Pallet-Type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6850

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2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF THE ARMY
Instruction Manual 136-300-93 - Volumeter For Gas Mask Canister Sorbents (Q-134).
Pamphlet No. 2, Part 1 - Canister Test Methods.
Section E - Carbon Monoxide.

(Application for copies should be addressed to Directorite, Quality Assurance, United States Army, Edgewood Arsenal, Maryland 21010, ATTN: Chief, Inspection Aids Division.)

3. REQUIREMENTS

3.1 Qualification. - The catalyst furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Particle size. - The catalyst shall conform to the requirements specified in table 1.

Table I - Particle size requirements.

Sieves[1]/	Percent by weight	
	Maximum	Minimum
Retained on number 4 sieve	2	---
Passing number 4, retained on number 6	70	50
Passing number 6, retained on number 8	50	30
Passing number 8 sieve	5	---

[1]/Sieves shall conform to Specification RR-S-366.

3.3 Moisture. - The moisture content shall not be more than 0.5 percent.

3.4 Activity. - The catalyst shall not permit the passage of more than 50 parts per million (p.p.m.) of carbon monoxide at 100° plus or minus 5°C.; nor 0.5 p.p.m. of ethylene at 205° plus or minus 5°C. during or at the end of 30 minutes of testing when tested in accordance with 4.5.4.1 and 4.5.4.2 respectfully.

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3.5 Apparent density. - The apparent density shall be between 1.00 and 1.10 grams per cubic centimeter (gm./c.c.).

3.6 Material. - The material shall consist of a mixture of copper oxide and manganese oxide.

3.7 Mixture. - Before packaging (see 5.1), the catalyst shall be mixed with lithium hydroxide in accordance with MIL-L-20213 to make up a composition by volume of 90 percent catalyst and 10 percent lithium hydroxide.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests. - Qualification tests shall consist of the tests specified in 4.5 and 4.6. The tests of 4.5 shall be conducted at a laboratory satisfactory to the Bureau of Ships. After completion of the tests specified in 4.5 and approval of the test report by the Bureau of Ships, the service test specified in 4.6 will be authorized. One-hundred-ten pounds of catalyst without the lithium hydroxide added, shall be provided for qualification tests.

Application for Qualification test shall be made in accordance with "Provisions Governing Qualification" (see 6.4 and 6.5).

4.3 Sampling. -

4.3.1 Lot. - A lot shall consist of all the catalyst produced, before mixing in the lithium hydroxide, and offered for delivery at one time.

4.3.2 Sampling for quality conformance tests. - Five specimens shall be selected as representative of each lot. The specimens may be taken from five randomly selected containers or at intervals throughout the container filling operation. The specimens shall be placed in clean, dry, airtight containers, and labeled to identify the container and lot represented. After determination of conformance with 4.4.2, all five specimens shall be subjected to the tests of 4.5.1 and 4.5.3. Three of the five specimens shall be subjected to the tests of 4.5.2 and 4.5.4.

4.3.3 Sampling for examination of filled containers. - Filled containers shall be selected in accordance with MIL-STD-105 at inspection level 1 for the examination specified in 4.4.1. The acceptable quality level shall be 2.5 percent.

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4.4 Examination. -

4.4.1 Examination of filled containers. - Each filled container selected in accordance with 4.3.3 shall be examined for leakage, defects of closure, and marking, and weighed to verify the fill. Then each container shall be emptied, examined for homogeneous mixing of the lithium hydroxide and the catalyst, refilled and returned to the shipment.

4.4.2 Examination for moisture content. - All specimens shall be examined for compliance with 3.3. Examination shall be stopped if any specimen fails to meet this requirement.

4.5 Test procedures. -

4.5.1 Particle size. - Sieves shall be selected conforming to the sizes specified in 3.2. The sieves shall be arranged in order of decreasing mesh size, placing the largest mesh on top and a pan on the bottom. One-hundred ± 0.5 grams (gm.) of the specimen shall be weighed and placed on the top sieve. A cover shall be placed on the top sieve and shall be assembled in a single-eccentric-type mechanical shaker geared to produce 300±10 taps of the striker per minute. The machine shall be operated for 2 minutes. The portion retained on each sieve shall be weighed and the results calculated in percentage.

4.5.2 Moisture. - Ten grams of the specimen shall be weighed to the nearest milligram (mg.), in a tared low-form weighing bottle. The specimen shall be heated for 1.5 hours at 150 ± 5°C. with the cover removed and shall be cooled in a desiccator and reweighed. The loss in weight shall be calculated as percent moisture.

4.5.3 Filling test container. - The volume of the specimen shall be measured in accordance with Instruction Manual 136-300-93. The specimen shall be dried in accordance with the method specified in 4.5.2. The bottom part of the container shall be filled as shown on figure 1 with 485 milliliter (ml) of specimen conforming to 3.2 by means of the filling tube shown on figure 2. Filling of the container shall be completed by adding an additional 485 ml of specimen using the filling tube. The container closure shall be placed in place on top of the bed and fastened securely. The test container shall be mounted in the flow system to affect an upward vertical flow.

4.5.4 Activity. -

4.5.4.1 Carbon monoxide test. - The filled container shall be placed in the test stream. Ten cubic feet per minute (c.f.m.) of dry air shall be passed with a dew point of minus 40°C., having a concentration of 1000 p.p.m. of carbon monoxide and a temperature of 100 ± 5°C. through the catalyst co tainer for 30 minutes.

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4.5.4.2 Ethylene test. - Immediately following the test specified in 4.5.4.1, the air temperature shall be raised and held at 205 ± 5°C. After 5 minutes, into the 10 c.f.m. of air flow, 40 to 50 p.p.m. of ethylene shall be introduced down-stream of the drier. The test shall be continued for 30 minutes.

4.5.4.3 Test analysis. - During the activity tests the effluent air from the test container shall be analyzed by a method capable of accurately determining carbon monoxide to 20 p.p.m. and ethylene to 0.5 p.p.m.

4.5.5 Apparent density. - To determine the apparent density, a 50 ml graduate cylinder (Fisher 8-549 or equal) shall be filled by supporting the cylinder and a funnel (Fisher 10-345 or equal) with a diameter of 60 millimeters (mm) and a stem diameter of 14 mm in a vertical position with the stem of the funnel centered in the cylinder. The catalyst shall be poured, a few grains at a time, onto the side of the funnel. The grains should not have a free fall through the apex of the funnel at any time. The cylinder shall be filled to the 50 ml mark. The weight of catalyst shall be determined and the apparent density shall be calculated.

4.6 Service test. - The catalyst shall be evaluated on board an operating submarine under prolonged operating conditions for a period of a minimum of thirty days with cycling of operation in accordance with current operating procedures, but no less than ninety-six hours of continuous operation per burner per cycle. After completion of the submarine service test, the catalyst will be returned to the Naval Research Laboratory for verification of its activity.

4.7 Rejection. - If any specimen fails to pass the examination and tests specified herein, this shall be cause for rejection of the lot.

4.8 Examination of preparation for delivery. - An examination shall be made to determine that packaging, packing and markings comply with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing. -

5.1.1 The material shall be packed in 5 gallon pails conforming to type 1, class 3 of PPP-P-704. Screw cap closures and inner seals shall be required. Exterior coating shall be required, color shall be olive drab. Wire handles or bails shall be galvanized or protectively coated to resist corrosion. Each pail shall be filled with an equal quantity of 30 pounds of material.

5.1.2 Pallets. - When specified (see 6.2), shipping containers shall be palletized in accordance with MIL-STD-147.

5.2 Marking. - In additon to any special marking required by the contract or order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. - The material covered by this specification is intended for use in the oxidation of carbon monoxide and hydrocarbons.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether palletization is required (see 5.1.2).

6.3 Batch. - A batch is defined as the quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

6.4 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-21665 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.5.).

6.5 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

6.6 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIRMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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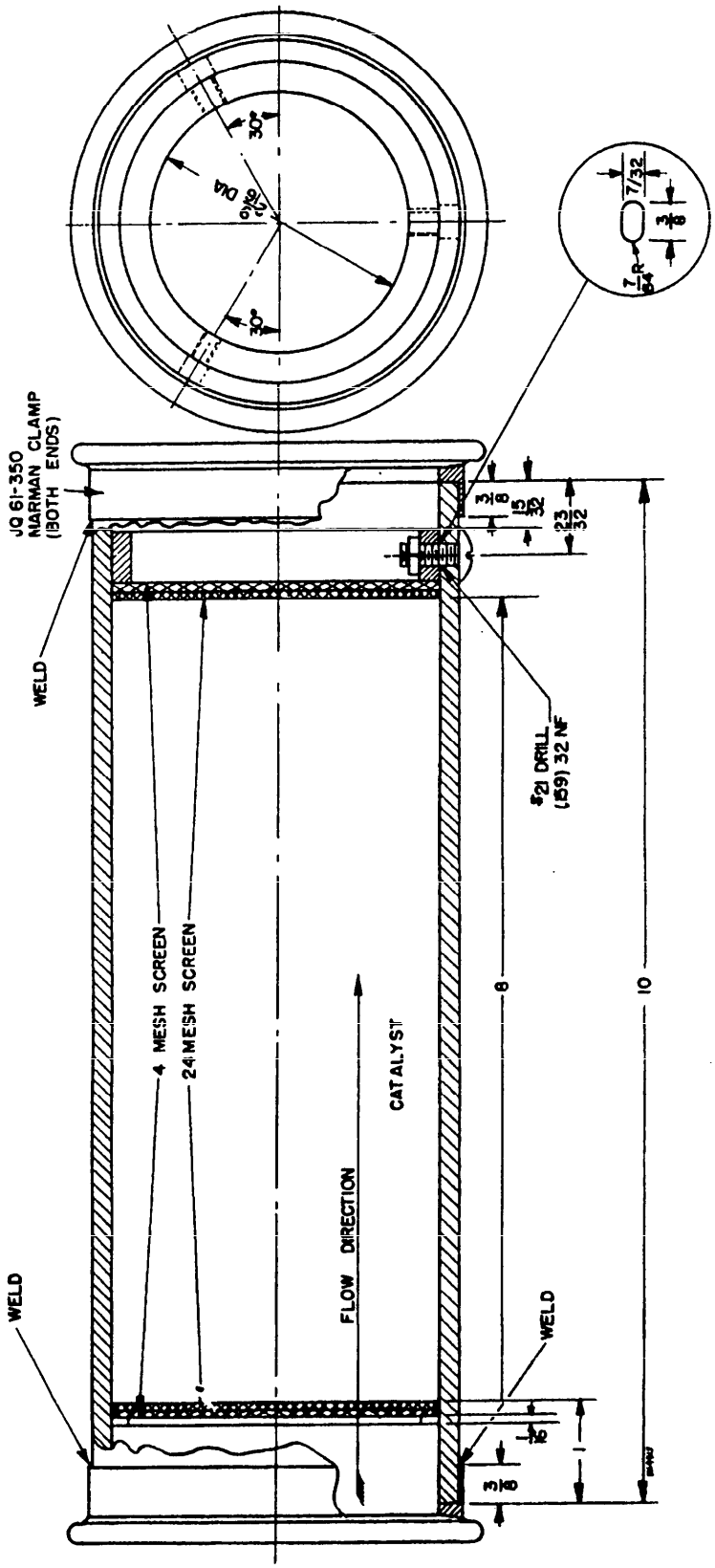


Figure 1 - Catalyst test cell.

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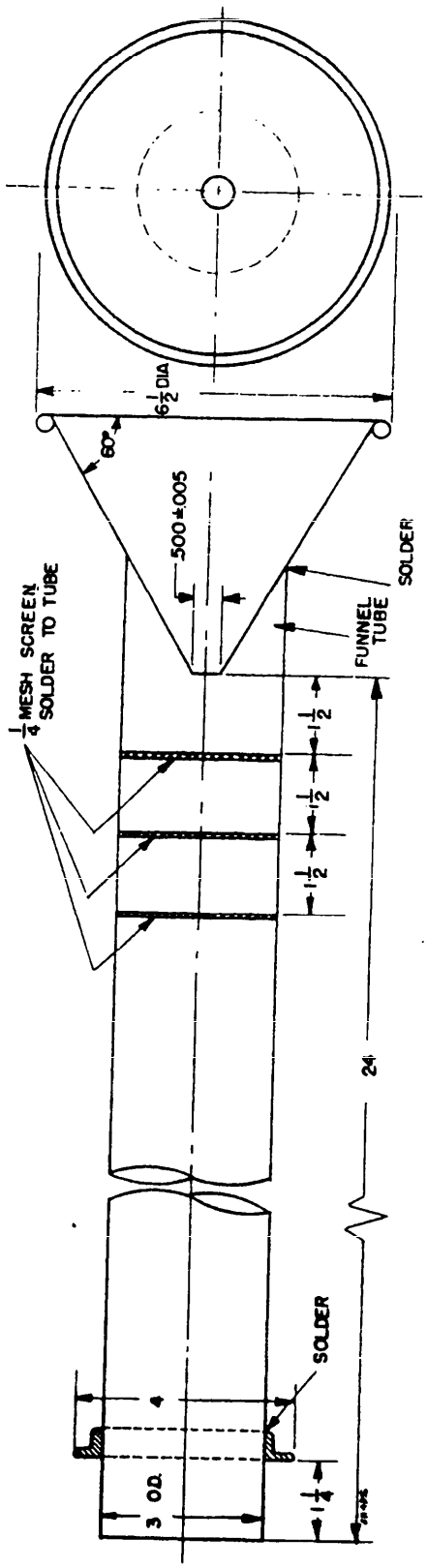


Figure 2 - Filling tube.

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